

Lab 1: Requirements Table

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Problem Statement #33 — Micro-Lending & Peer Credit Risk Assessor

Actors: Borrower, Peer Lender

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier (Low/Med/High) accurately assigned. Fail: Incomplete financial profile generates credit approval.	Risk-based decisioning is the core value proposition of the platform; lenders and borrowers both depend on an accurate tier.
FR-002	Functional	The system shall allow a borrower to submit a loan application specifying requested amount, tenure, and purpose, along with supporting financial details.	High	Pass: Application is saved with a unique ID and status "Pending Review". Fail: Application with a missing mandatory field is accepted for review.	A structured application is the entry point for risk scoring and lender matching.
FR-003	Functional	The system shall generate a structured EMI (Equated Monthly Installment) repayment schedule for an approved loan, based on principal, risk-adjusted interest rate, and tenure.	High	Pass: Sum of all scheduled EMIs equals principal plus interest, within rounding tolerance of Rs.1. Fail: Generated schedule total deviates from the expected amount.	Both borrower and lender need clear, verifiable visibility into repayment obligations and timelines.
FR-004	Functional	The system shall automatically trigger a default warning notification to the borrower and the peer lender when a scheduled EMI is not received within a defined grace period.	Medium	Pass: Notification is dispatched to both parties within 24 hours of grace-period expiry. Fail: A missed EMI produces no notification after the grace period.	Early default warnings allow lenders to react and reduce platform-wide credit risk.
FR-005	Functional	The system shall allow a peer lender to browse open loan listings filtered by risk tier and commit funds toward one or more listings.	Medium	Pass: Lender's committed amount updates the loan's funding progress in real time and cannot exceed the requested amount. Fail: Total committed funds exceed the requested loan amount.	Matching lenders to borrowers by risk appetite is the core marketplace function.
NFR-001	Non-functional (Performance & Security)	Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Any concurrent transaction test produces a double-spend or partial commit.	Financial correctness under concurrent access is non-negotiable for a lending platform.
NFR-002	Non-functional (Security)	The system shall encrypt all borrower financial and personal data at rest and in transit using industry-standard encryption (e.g., AES-256 at rest, TLS 1.2+ in transit).	High	Pass: Security audit/log review confirms no sensitive field is stored or transmitted in plaintext. Fail: Any sensitive financial field is found unencrypted at rest or in transit.	Protects sensitive financial/PII data and supports regulatory compliance for a fintech platform.